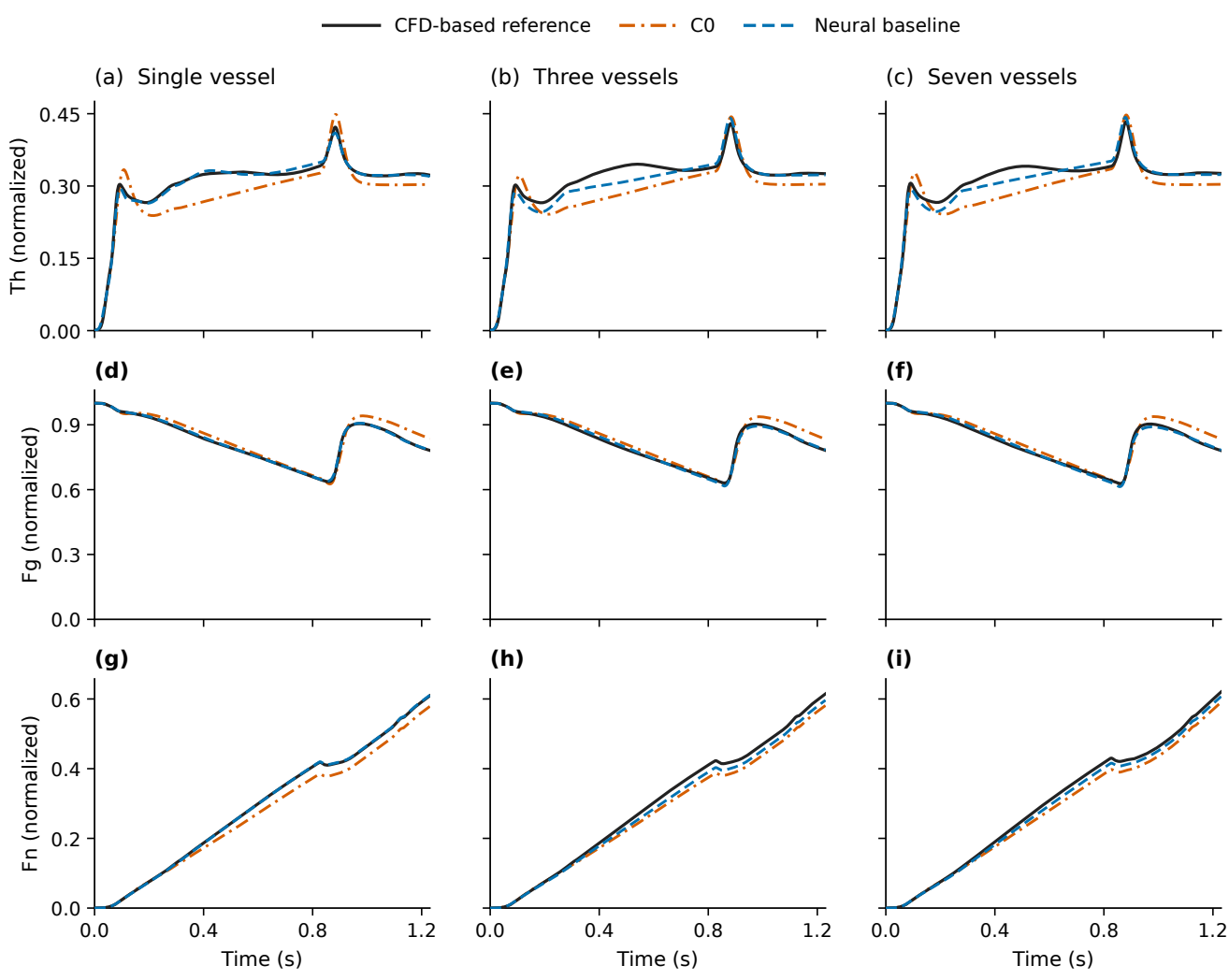
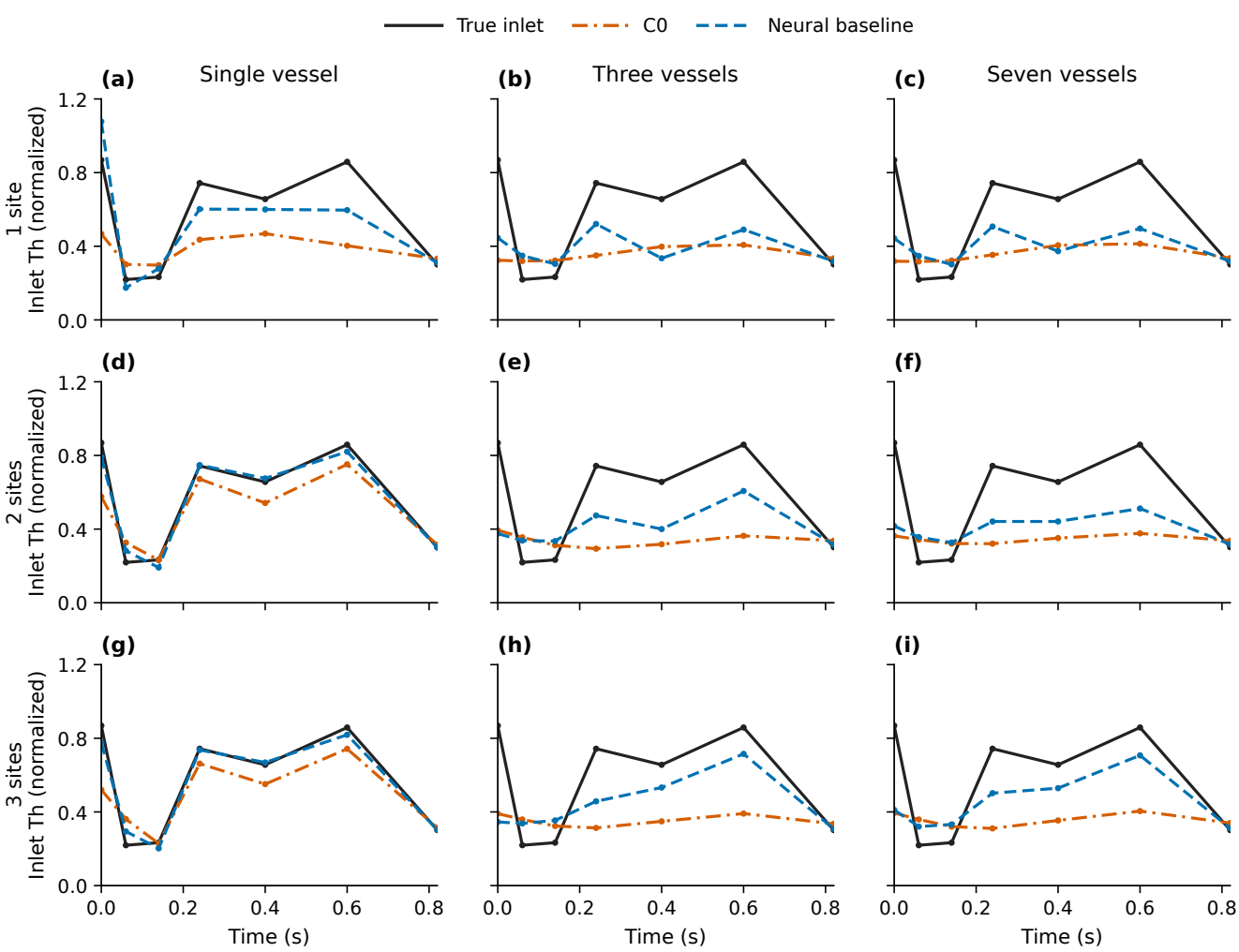
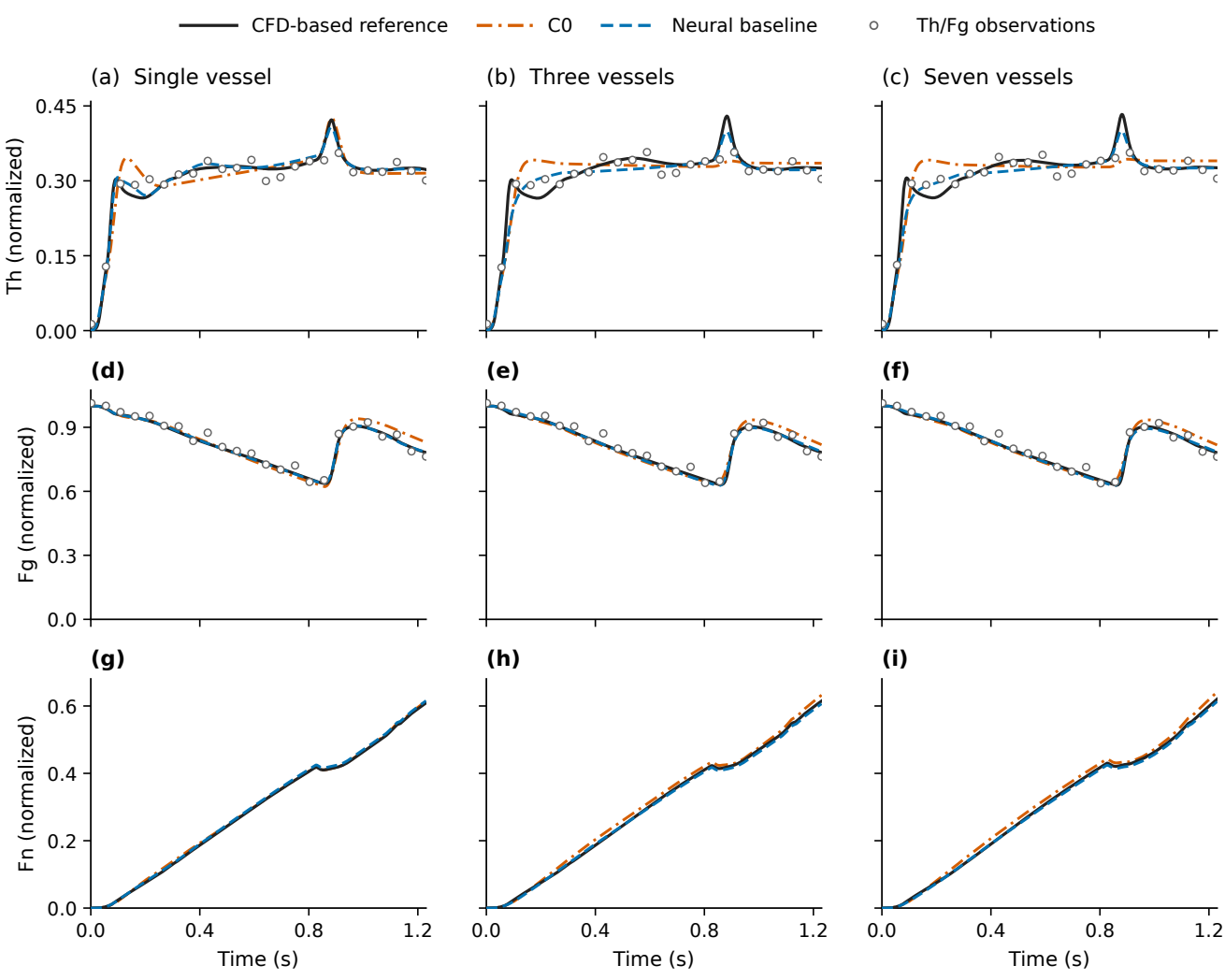
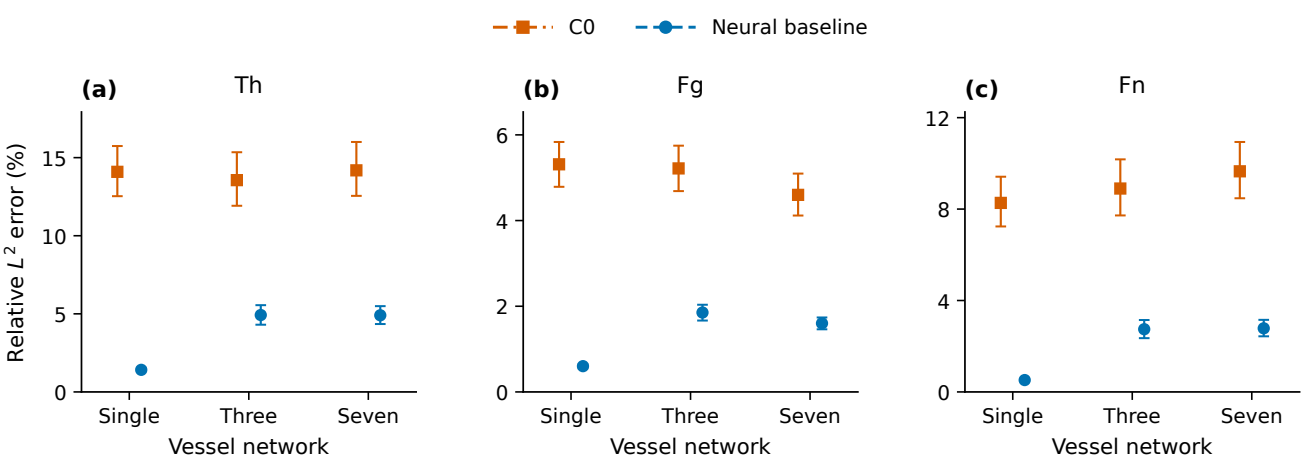


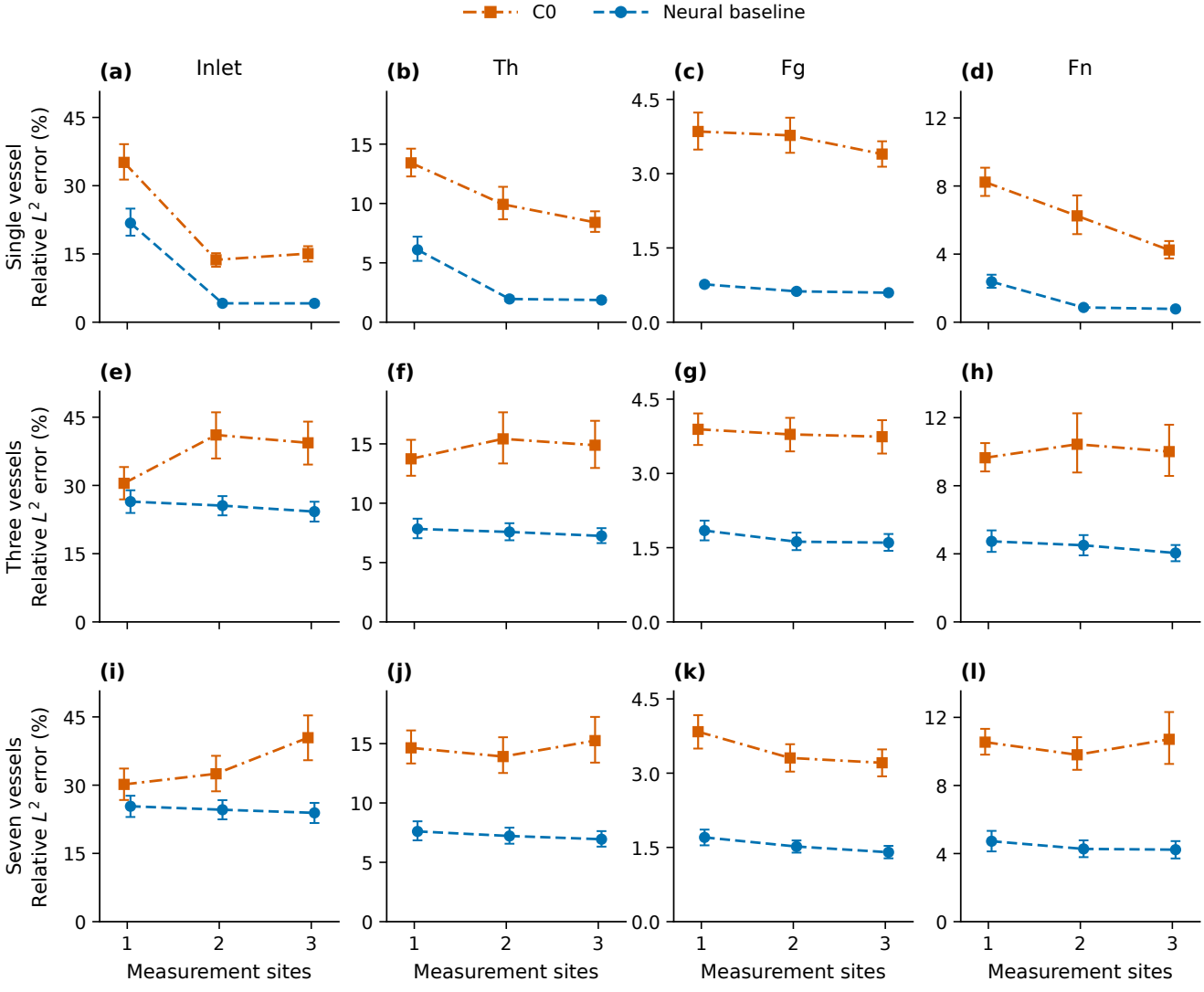
Test case 80; neural seed 42; Known inlet

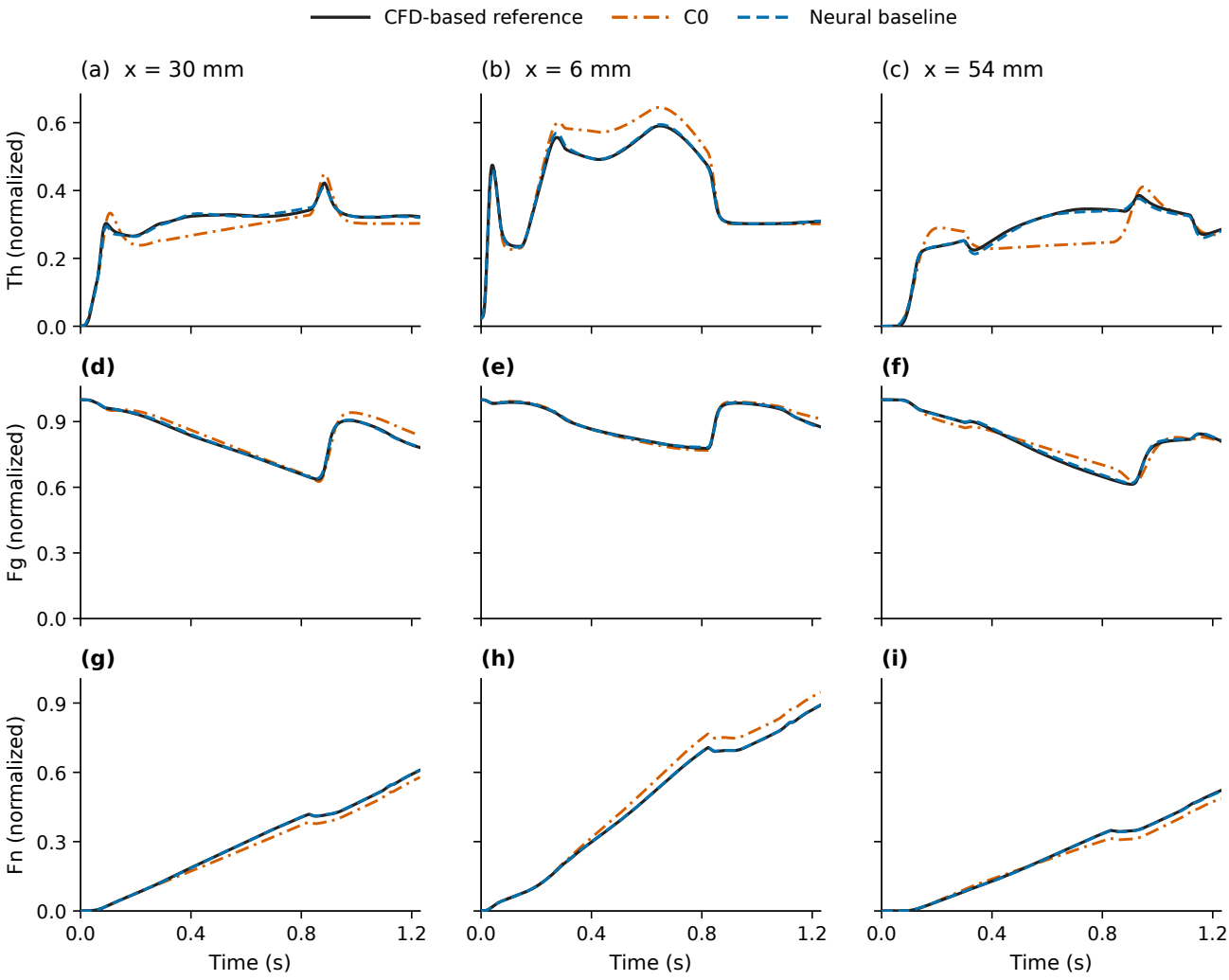


Test case 80; neural seed 42; noise realization 1

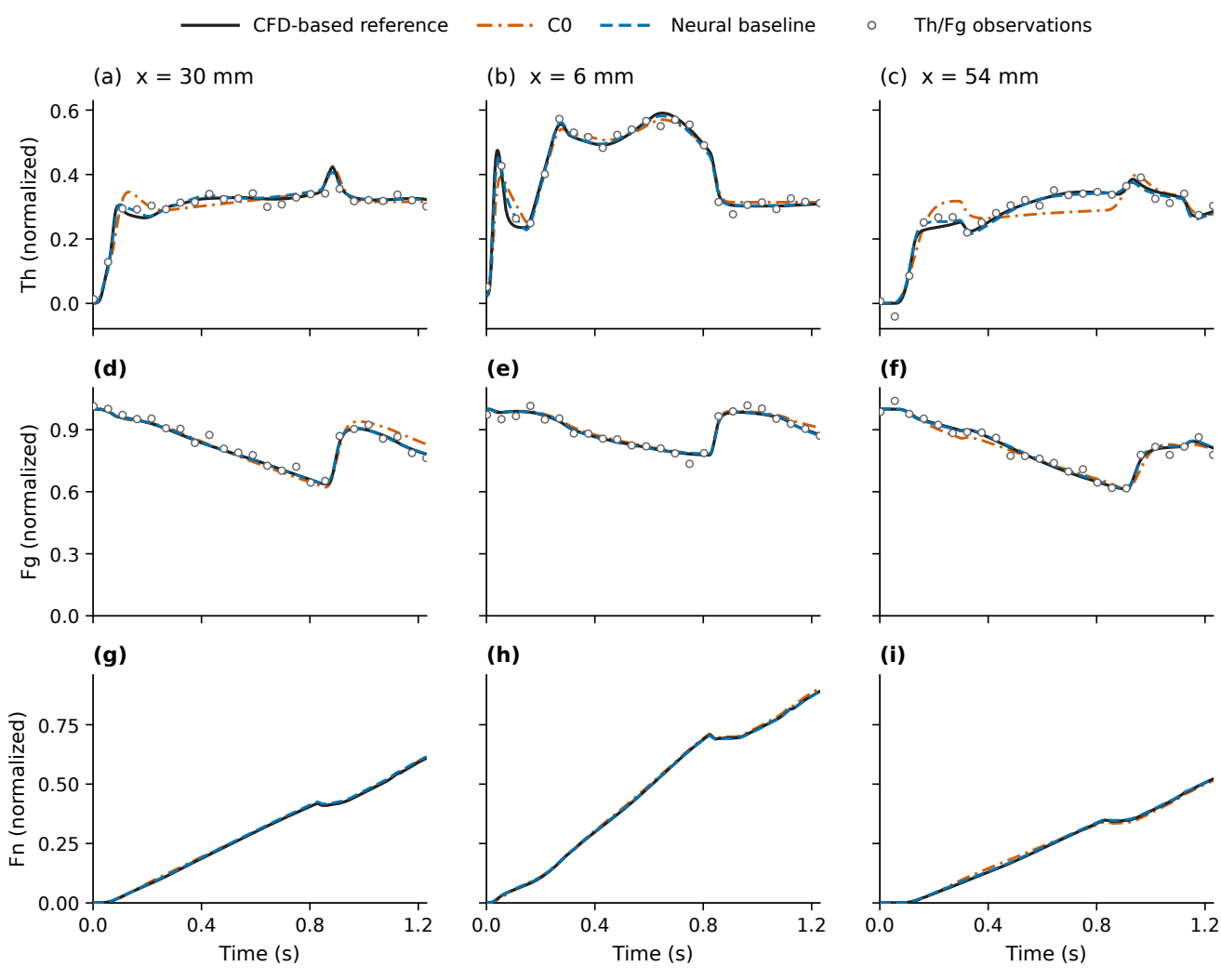

CFD-based reference
C0
Neural baseline
Th/Fg observations
(a) Single vessel
(b) Three vessels
(c) Seven vessels
Th (normalized)
Fg (normalized)
Fn (normalized)
(d)
(e)
(f)
(g)
(h)
(i)
0.45
0.30
0.15
0.00
0.9
0.6
0.3
0.0
0.6
0.4
0.2
0.0
0.0
0.4
0.8
1.2
Time (s)
Test case 80; neural seed 42; 3 sites; noise realization 1

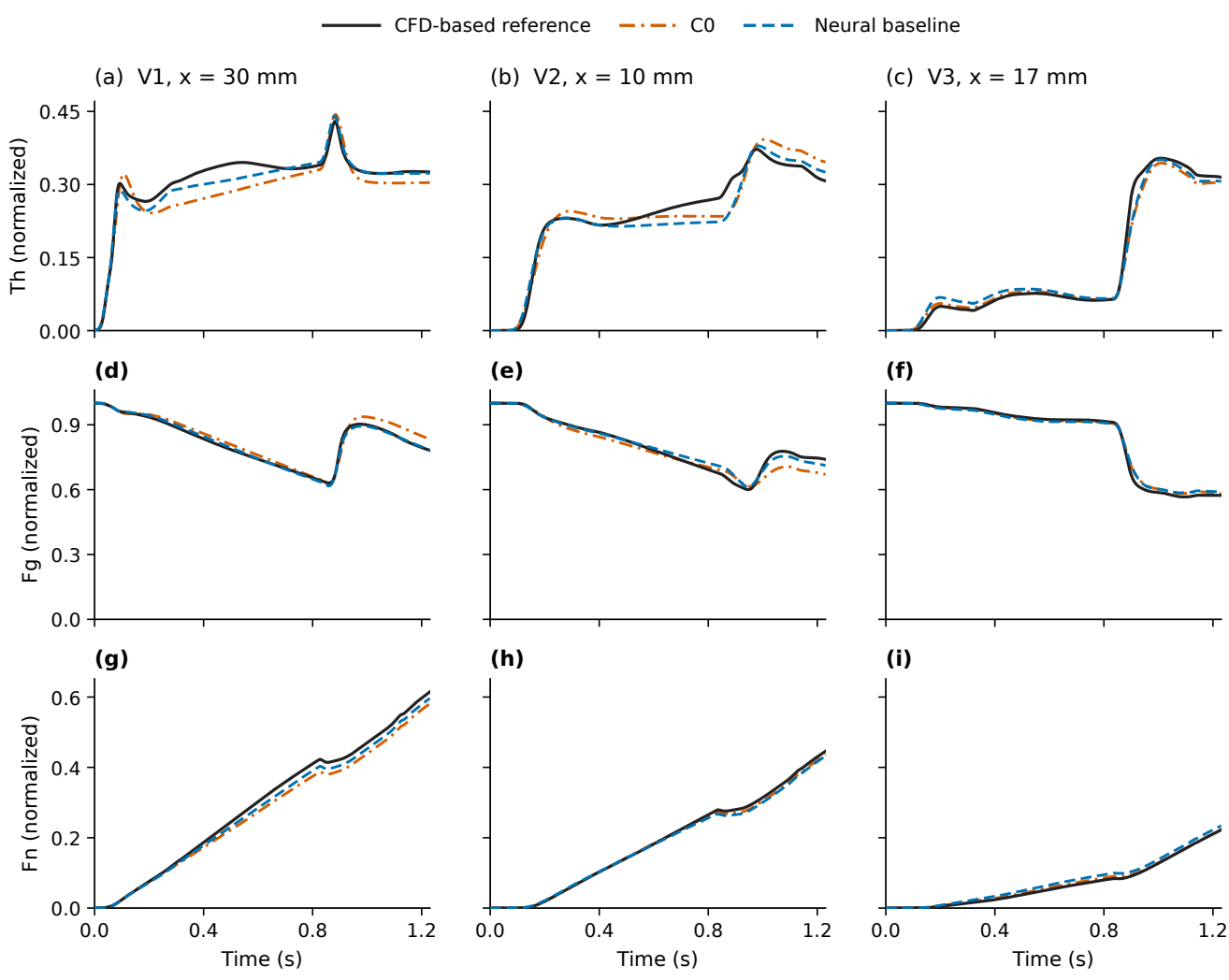

C0
Neural baseline
(a)
Th
(b)
Fg
(c)
Fn
Relative L^2 error (%)
0
5
10
15
0
2
4
6
0
4
8
12
Single
Three
Seven
Vessel network
32 test inlet cases; mean and 95% case-bootstrap CI

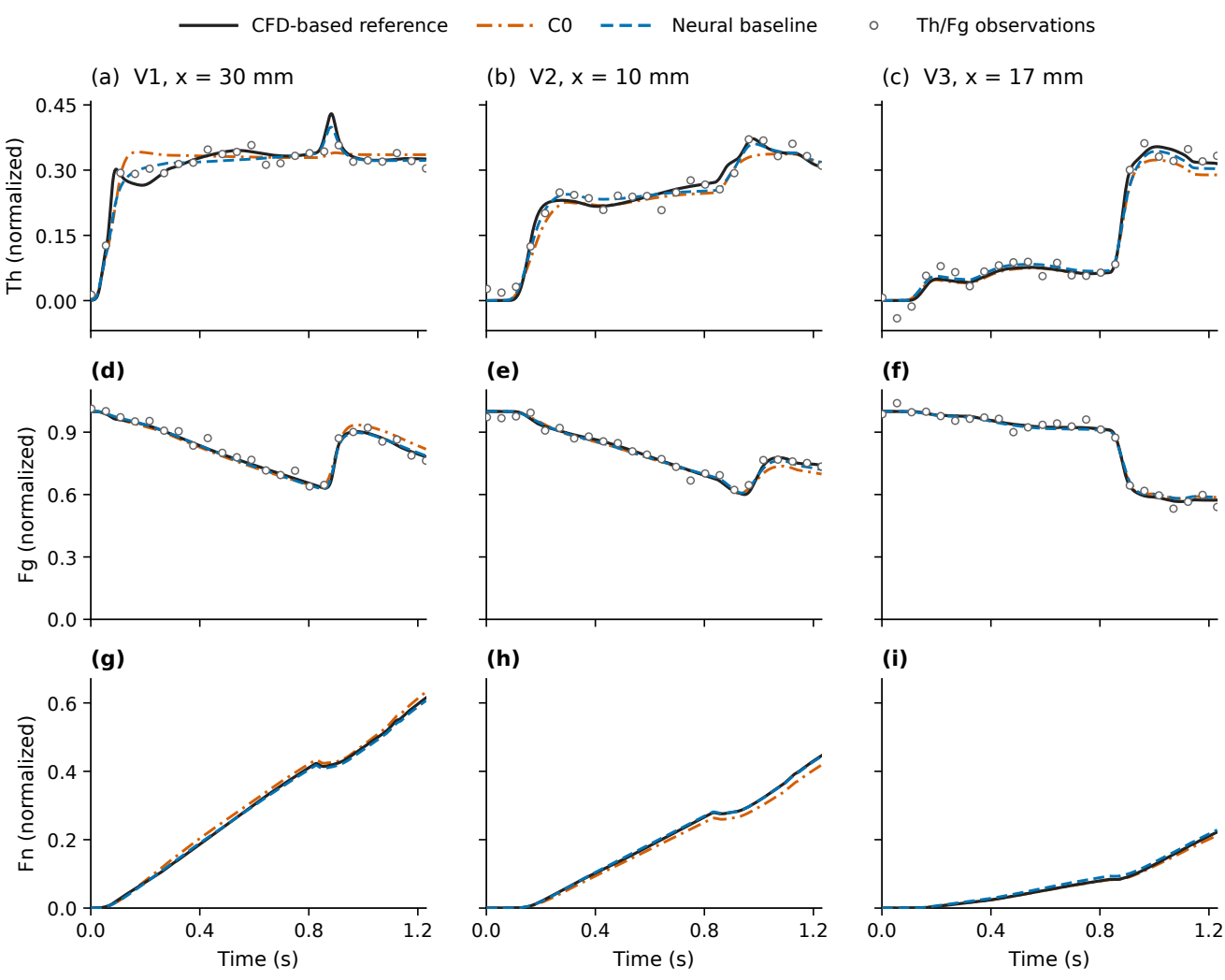

C0
Neural baseline
(a) Inlet
(b) Th
(c) Fg
(d) Fn
(e)
(f)
(g)
(h)
(i)
(j)
(k)
(l)
Single vessel
Three vessels
Seven vessels
Relative L^2 error (%)
Measurement sites
32 test inlet cases; mean and 95% case-bootstrap CI

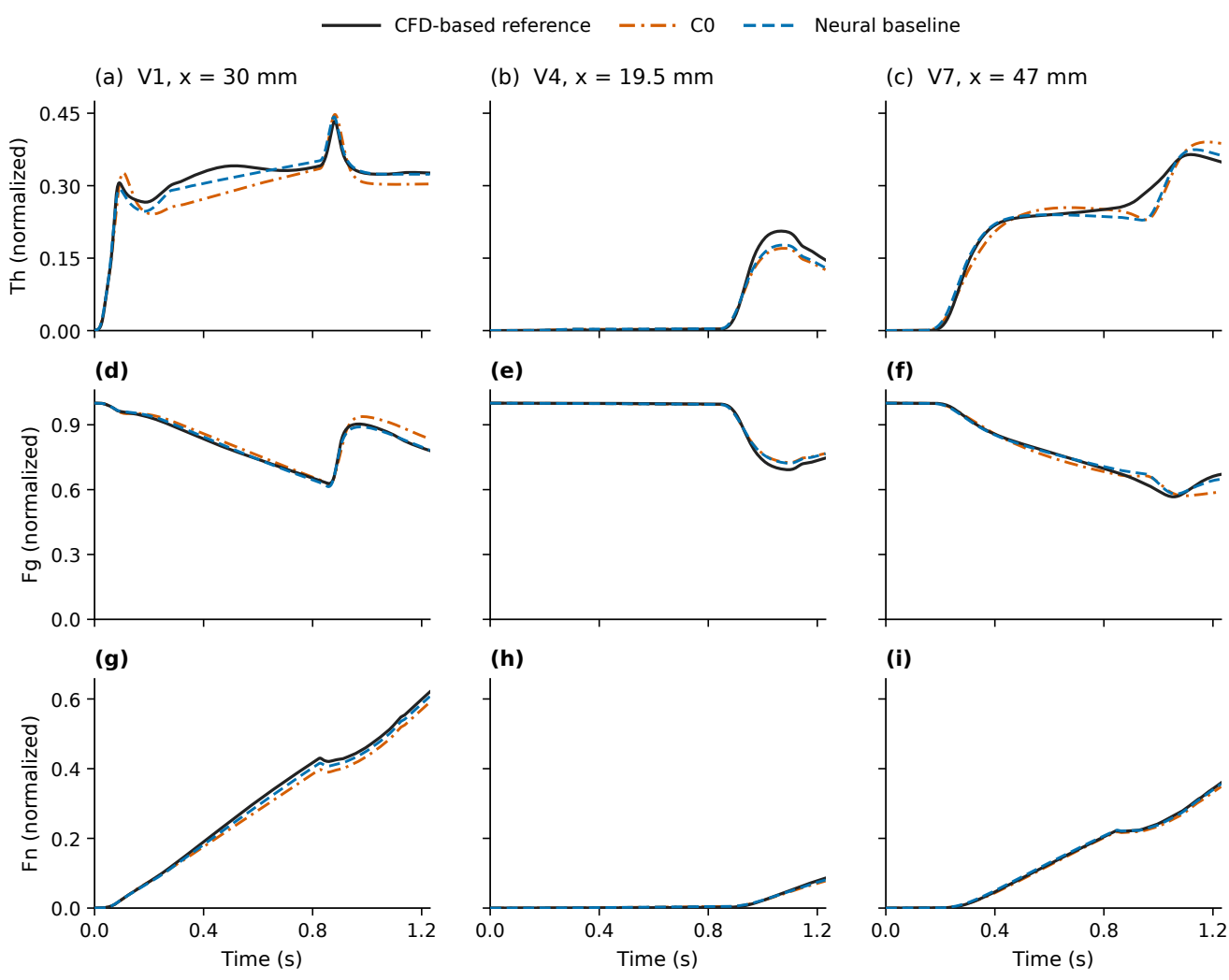


Test case 80; neural seed 42; Known inlet

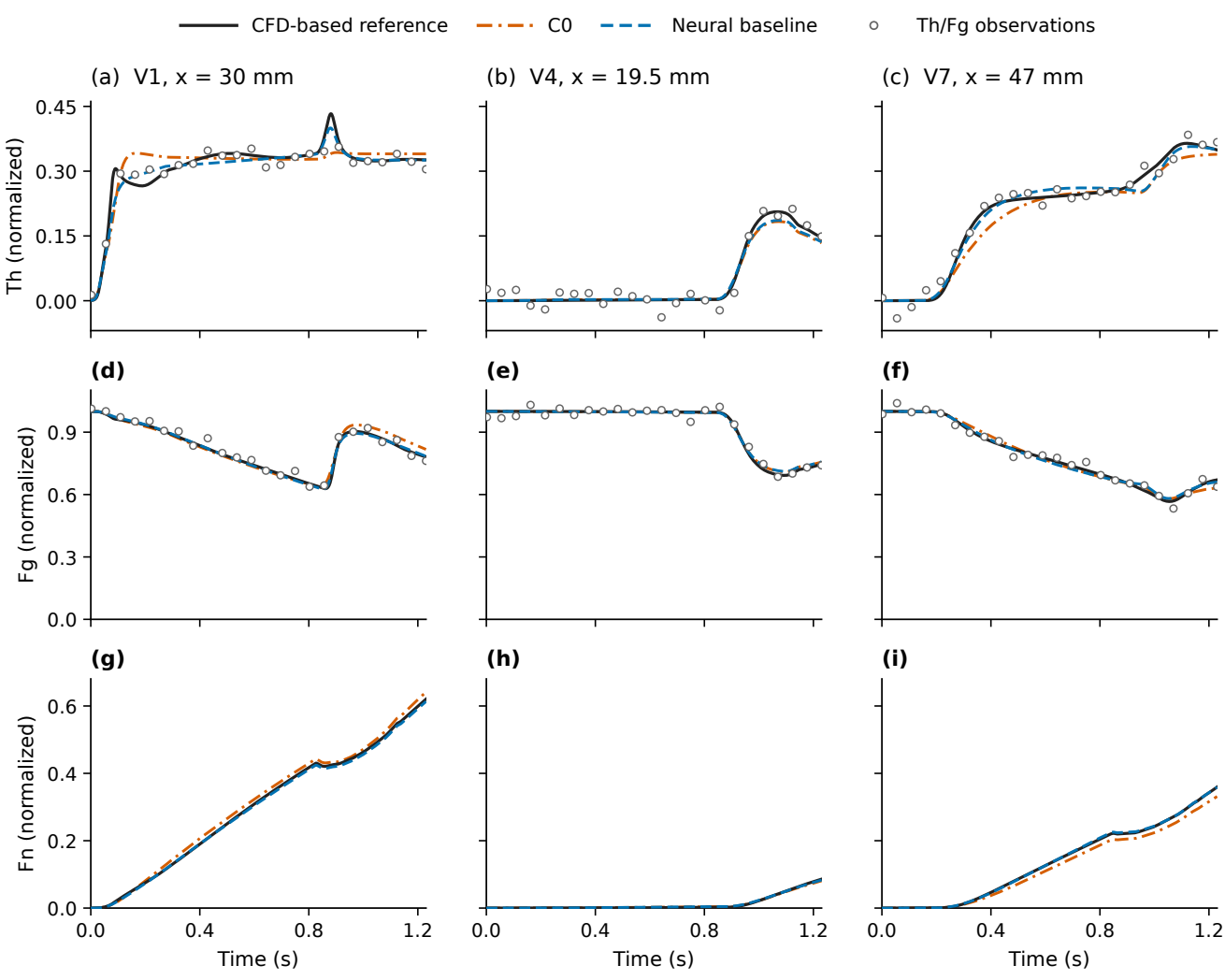

CFD-based reference
C0
Neural baseline
Th/Fg observations
(a) x = 30 mm
(b) x = 6 mm
(c) x = 54 mm
Th (normalized)
Fg (normalized)
Fn (normalized)
(d)
(e)
(f)
(g)
(h)
(i)
Time (s)
Time (s)
Time (s)
Test case 80; neural seed 42; 3 sites; noise realization 1


CFD-based reference
C0
Neural baseline
(a) V1, x = 30 mm
(b) V2, x = 10 mm
(c) V3, x = 17 mm
Th (normalized)
0.45
0.30
0.15
0.00
(d)
(e)
(f)
Fg (normalized)
0.9
0.6
0.3
0.0
(g)
(h)
(i)
Fn (normalized)
0.6
0.4
0.2
0.0
0.0
0.4
0.8
1.2
Time (s)
Test case 80; neural seed 42; Known inlet


CFD-based reference
C0
Neural baseline
Th/Fg observations
(a) V1, x = 30 mm
(b) V2, x = 10 mm
(c) V3, x = 17 mm
Th (normalized)
0.45
0.30
0.15
0.00
(d)
(e)
(f)
Fg (normalized)
0.9
0.6
0.3
0.0
(g)
(h)
(i)
Fn (normalized)
0.6
0.4
0.2
0.0
0.0
0.4
0.8
1.2
Time (s)
Test case 80; neural seed 42; 3 sites; noise realization 1



Test case 80; neural seed 42; Known inlet



Test case 80; neural seed 42; 3 sites; noise realization 1